
Fixed-Spectrum Swaps Separate Attention Capacity from Assignment

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Abstract

1 Interpretability often reads attention mass on a gold token as evidence that a
2 transformer routes information to it. We show that this inference is unsafe in a
3 specific, testable way. Statistics of the attention weights alone, including entropy,
4 the maximum, every norm, and the sorted spectrum, are invariant to which token
5 holds which weight, so they cannot identify assignment; and attention mass on the
6 gold token, which does identify assignment, still does not determine the causal
7 effect of that assignment on the output. We introduce the fixed-spectrum swap
8 (FSS), a causal intervention that preserves every permutation-invariant property of
9 an attention row while changing its assignment to task-labeled positions. Across
10 controlled retrieval, pretrained language models, and context-grounded question
11 answering, FSS separates selective capacity from correct assignment and reveals
12 assignment-responsive, saturated, and selector-mismatch regimes. Selecting cir-
13 cuits by source mass fails where selecting by a local value-aware utility term
14 succeeds: source-mass selection produces near-zero held-out margin change, while
15 utility selection is large and positive and ranks finite margin effects with per-model
16 Spearman correlations of 0.54 to 0.87. Context-length sweeps mark where the di-
17 agnosis stops being sufficient, since behavior can degrade without a corresponding
18 routing change. FSS gives a controlled test of whether a specified post-softmax
19 reassignment of attention changes a model’s prediction.

20 1 Introduction

21 Interpretability claims about transformers often rest on how attention is distributed. Attention weights
22 are read as an explanation of which tokens a prediction depends on [Jain and Wallace, 2019, Wiegrefe
23 and Pinter, 2019], and attention mass on a gold token, or a summary of it such as low entropy or a
24 large maximum, is taken as evidence that the model routes information to that token. We ask whether
25 that inference is licensed, and we separate two versions of it. Summary statistics of the weights
26 alone cannot even identify assignment; attention mass on the gold token can, yet still may not tell us
27 whether the routing causally drives the prediction.

28 The first gap is structural. Entropy, the maximum weight, every norm, and the full sorted spectrum
29 are invariant to permutations of token positions, so two attention rows can share all of them while one
30 places its largest weight on the required evidence and the other on a distractor. A measurement equal
31 on both rows cannot distinguish them. Attention mass on the gold token, m_S , is the sharper and more
32 common measure: it is not permutation-invariant and does pin down the assignment. Our question is
33 therefore not about m_S as a descriptor but about its causal warrant. At a fixed spectrum, does moving
34 mass onto the task-required evidence change the prediction, and is m_S enough to say when it will?

35 We test this with a fixed-spectrum swap (FSS). At the answer query, we exchange the weight on a
36 known source with a distractor weight and then run the unchanged model. Because the exchange

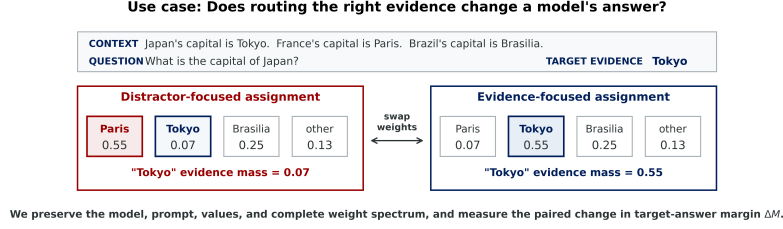


Figure 1: Question-answering use case. For the same context and question, a fixed-spectrum swap (FSS) moves the maximum attention weight from a distractor (“Paris”) to required evidence (“Tokyo”). The model, values, and weight spectrum remain unchanged; the paired target-answer margin tests whether assignment affects the answer.

preserves the complete spectrum, a distractor-only exchange gives a matched control. The answer depends on the spectrum’s selective capacity, its assignment to task-labeled positions, and the downstream utility of the moved values, and we find that source mass alone does not determine it: selecting circuits by source mass fails where selecting by a value-aware utility term succeeds.

Length is a boundary we test, not a phenomenon we promise to explain. Performance often declines as distractors accumulate [Kazemnejad et al., 2023, Wang et al., 2024, Vasylenko et al., 2025], and attention grows more diffuse while its output variance can collapse [Li et al., 2025]. We use context-length sweeps to ask how far the fixed-spectrum diagnosis remains informative, and we report where it stops rather than claiming it accounts for long-context failure.

Our contributions are:

- **Fixed-spectrum identification.** We separate selective capacity from task-conditioned assignment by preserving the complete attention spectrum and altering only its token placement.
- **Source mass is not sufficient.** We show that source mass identifies assignment but does not determine the causal effect: selecting circuits by source mass fails where a value-aware utility term succeeds.
- **Transfer and boundaries.** We test the intervention across controlled tasks, pretrained models, multi-evidence computation, and natural QA, and identify saturation and selector-mismatch regimes together with the endogenous and length boundaries where the diagnosis stops being sufficient.

2 Related Work

Attention weights as routing evidence. Attention weights are widely read as evidence of which tokens a prediction uses, a practice examined in the attention-as-explanation debate [Jain and Wallace, 2019, Wiegrefe and Pinter, 2019, Brunner et al., 2020]. Concentration summaries such as entropy, maximum weight, and output variance are connected to optimization, token selection, and extrapolation [Tarzanagh et al., 2023, Wang et al., 2024, Nakanishi, 2025, Vasylenko et al., 2025, Li et al., 2025, Zsámboki et al., 2025], and move together as context length changes. Our target is not these summaries, which few defend as routing measures, but attention mass on the gold token, which is the measure such claims actually rely on. Mass identifies assignment, so the open question is causal: does that assignment drive the output, and is mass enough to predict when. We test this while the entire spectrum is held fixed.

Attention intervention and non-identifiability. Attention weights have long been studied as explanations, perturbed to test prediction sensitivity, and shown to be non-identifiable without their value vectors [Jain and Wallace, 2019, Wiegrefe and Pinter, 2019, Brunner et al., 2020, Kobayashi et al., 2020, Hassid et al., 2022]. Causal circuit analysis patches internal activations and measures the resulting output change [Yamakoshi et al., 2023, Conmy et al., 2023]. Attribution patching uses a gradient approximation to screen many candidate components efficiently [Syed et al., 2023, Kramár

et al., 2024]. Our utility term has a related first-order form. Here we evaluate it along an exact permutation of a model’s own attention row, alongside the finite swap itself.

Closest attention interventions. The nearest methods delete, block, renormalize, or rank parts of an attention pathway: Attention Knockout blocks selected edges and renormalizes the row [Geva et al., 2023], efficient attention projects a row onto its output-effective subspace [Naim and Asher, 2024], and Contribution Weights rank tokens by attention mass combined with projected-value magnitude, testing importance through removal [Cunningham and Muca Cirone, 2026]. Each changes the weight multiset. Our reassignment leaves it intact and changes only its assignment to token positions, and Equation 5 connects that finite, task-labeled swap to the target margin. Sparse and retrieval-based architectures instead change how distant context is accessed [Vasylenko et al., 2025, Hu et al., 2025]; we study a trained row with a fixed spectrum and isolate the effect of assigning its available selectivity to task-required evidence.

3 Methods

3.1 Capacity and Assignment

At an answer query, let $a = (a_1, \dots, a_n)$ be an attention row over n token positions, where $a_j \geq 0$ and $\sum_j a_j = 1$. Let $S \subseteq \{1, \dots, n\}$ contain the $r = |S|$ task-required evidence positions, and let $a_{(1)} \geq \dots \geq a_{(n)}$ denote the same weights in descending order. Define

$$C_r^+ = \sum_{i=1}^r a_{(i)}, \quad C_r^- = \sum_{i=n-r+1}^n a_{(i)}, \quad (1)$$

$$K_r(a) = C_r^+ - C_r^-.$$

C_r^+ and C_r^- are the largest and smallest evidence masses available under permutations of the fixed spectrum, and $K_r(a)$ is the available routing range. An assignment is a permutation π of $\{1, \dots, n\}$ that places weight $a_{\pi(j)}$ at token position j . Its evidence mass and normalized assignment coordinate are

$$m_S(\pi) = \sum_{j \in S} a_{\pi(j)}, \quad (2)$$

$$\rho_S(\pi) = \frac{m_S(\pi) - C_r^-}{K_r(a)} \in [0, 1], \quad K_r(a) > 0.$$

Thus $m_S(\pi) = C_r^- + \rho_S(\pi)K_r(a)$. If $K_r(a) = 0$, every assignment has the same evidence mass.

Capacity specifies how much evidence mass the spectrum can support.
Assignment specifies where that mass is placed.
Utility specifies whether moving it changes the answer.

Proposition 1 (Fixed-spectrum capacity-assignment decomposition). *For every assignment π of a fixed weight multiset,*

$$C_r^- \leq m_S(\pi) \leq C_r^+, \quad (3)$$

and both bounds are attained. If $K_r(a) > 0$, Equation 2 defines a unique $\rho_S(\pi)$. No statistic invariant to token permutation can determine ρ_S .

Proof sketch. An r -position evidence set receives at most the r largest weights and at least the r smallest. Assigning those weights to S attains each endpoint. Normalizing evidence mass within this interval gives ρ_S . Any permutation-invariant statistic is constant across all assignments, including assignments at opposite endpoints. The supplementary material gives the full exchange argument and the corresponding affine margin range. The decomposition is algebraic for every r . Behavioral validation is positive for $r = 1$ and $r = 2$; the $r = 3$ interval includes zero and the $r = 4$ models miss competence, so efficacy at higher arity is unestablished.

3.2 Utility-Conditioned Output Effects

Capacity and assignment describe attention mass. Prediction also depends on values and downstream computation. Let H be the selected attention heads. In each $h \in H$, let s be the evidence position, d_h the donor position whose weight is exchanged with it, and a_{hj} the original attention weight on position j . Define the transferred mass $\delta_h = a_{hd_h} - a_{hs}$. Let x_{hj} be token j 's residual-stream contribution after head h 's value and output projections. If z is the unpatched residual state, the intervention gives $z' = z + \Delta z$, where

$$\Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h}). \quad (4)$$

Let $\ell_k(z)$ be the downstream logit for token k , with target y and competitor c . Define $M_c(z) = \ell_y(z) - \ell_c(z)$, the exact change $\Delta M_c = M_c(z + \Delta z) - M_c(z)$, and local utility $u_{hj}^{(c)} = \nabla_z M_c(z)^\top x_{hj}$.

Proposition 2 (Utility-conditioned swap effect). *If $\nabla_z M_c$ is L_c -Lipschitz on $\{z + t\Delta z : t \in [0, 1]\}$, then*

$$\Delta M_c = \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) + R, \quad |R| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \quad (5)$$

For an affine downstream margin, $R = 0$.

The Taylor remainder is standard. The new component is the task-labeled capacity-assignment coordinate coupled to the exact FSS vector in Equation 4.

The intervention transfers attention mass from a donor token to the source. It helps when the source contributes more positively to the target margin than the displaced donor, unless downstream curvature overwhelms that gain. Source-max can therefore succeed, saturate, or fail under the same attention-mass rule. The supplementary material derives Equation 5, extends it to evidence sets, and states a sufficient margin certificate.

4 Experimental Setup

Controlled tasks. Argmax retrieval presents key-value pairs and requests the value paired with the largest key. Flag retrieval presents values with exactly one marked position and requests that value. Both have one labeled source. Two-evidence modular sum marks two digits and requests their sum modulo ten, so both sources are required. Four-layer decoders are trained with NoPE and RoPE and four seeds per cell; evaluation reaches $50\times$ training length. An eight-layer, width-512 grid checks scale. The inverse temperature $\beta \in \{1, 2, 4\}$ changes attention capacity while each assignment comparison preserves the resulting sorted spectrum.

Pretrained and natural-language tasks. In-context key-value recall lists pairs followed by a query key; the paired value is the labeled source. We test Qwen2.5, Pythia, Gemma, and SmoLM2 across increasing pair counts. The selector audit compares circuits ranked on disjoint calibration data by source mass or by output-conditioned utility. Natural MCQA uses four-choice questions with one gold passage and unrelated passages. Questions must be correct with the full context and gold passage alone, but incorrect without context. This filtering defines a model-specific context-dependent subset and is not intended to estimate average performance over the full dataset. Its length ladder freezes the four-passage circuit before forming nested 8-, 16-, and 32-passage contexts with additional unrelated passages.

Interventions and estimation. The distractor control swaps the largest and smallest non-source weights, preserving source mass while permuting the same attention row. For a source position s , let $d^+ = \arg \max_j a_j$, $d^- = \arg \min_j a_j$, and $D = \{j : j \neq s\}$ over valid tokens. Source-max swaps (s, d^+) and source-min swaps (s, d^-) . In controlled and pretrained key-value recall, the distractor control uses the maximum and minimum within D . Natural QA uses a displacement-matched control. With $\delta = a_{d^+} - a_s$, it chooses

$$(p, q) = \arg \min_{\{j, k\} \subset D} |a_j - a_k| - \delta \quad (6)$$

151 and swaps (p, q) , minimizing the L_1 -displacement mismatch to the treatment while holding source
 152 mass fixed; every control preserves the complete spectrum. The control does not match the donor to-
 153 ken, positional distance, or value norm, which remain separate confounds. Calibration and evaluation
 154 examples are disjoint, circuits are frozen before paired evaluation, hierarchical intervals resample
 155 calibration or model seeds above paired examples, and a cell is interpreted only after its frozen
 156 competence gate passes. Three confirmatory contrasts form one inferential family (controlled source-
 157 max versus distractor control, the capacity-by-assignment interaction, and natural utility-selected
 158 source-max versus matched control), tested with two-sided exact sign-flip tests at the cluster level
 159 with Holm correction; other intervals are descriptive and unadjusted. These swaps are counterfactual
 160 probes and need not lie on the model’s endogenous query-key manifold. Treatment and control share
 161 this intervention class, so their contrast measures downstream sensitivity to assignment, rather than
 162 the frequency with which the model naturally creates a particular row. Layer controls (+1.107 for se-
 163 lected $K = 8$ versus -0.059 for a random layer) and coverage dose response show greater sensitivity
 164 under this probe. They do not show that normal query-key computation produces swapped rows or
 165 that selected heads are the exclusive retrieval mechanism. Section 5.4 reports the endogenous-row
 166 test and the patching comparison that bound this reading.

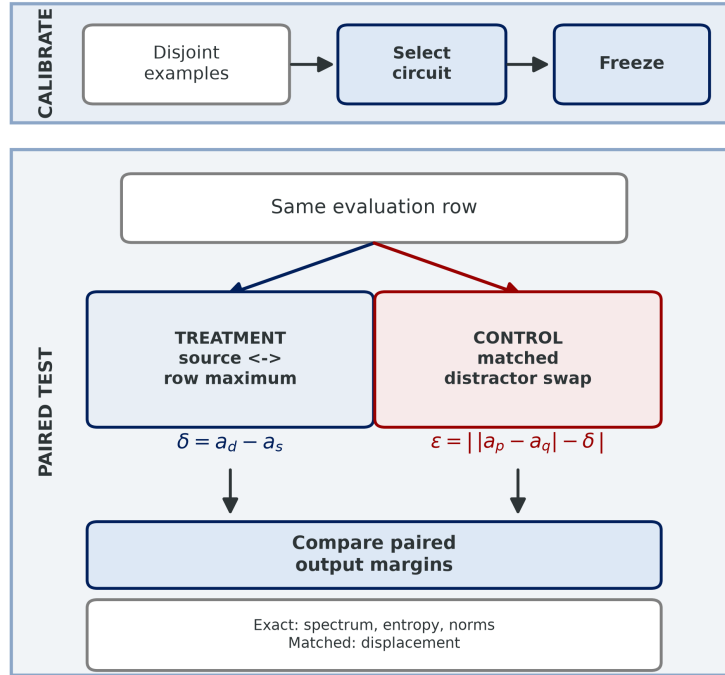


Figure 2: Paired causal protocol. A circuit selected on disjoint calibration data is frozen before treatment and control are applied to the same evaluation row. Both paths preserve the spectrum; the natural QA control additionally minimizes displacement mismatch in Equation 6.

167 **Measured outcomes.** We record task accuracy, target-versus-competitor margin, source mass,
 168 attention-output variance, entropy, norms, and the full sorted spectrum. Every fixed-spectrum
 169 condition preserves the selected attention rows to numerical precision. The supplementary material
 170 reports prompts, hyperparameters, per-length tables, competence failures, and preregistered decision
 171 rules.

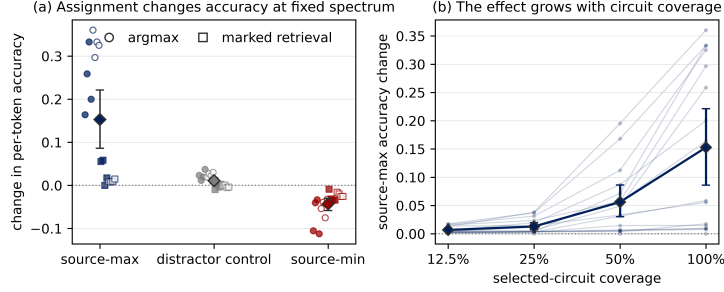


Figure 3: Assignment changes retrieval despite an identical attention spectrum. Source-max improves accuracy, source-min harms it, and the distractor-only control remains near zero (a). The effect increases with the fraction of the selected circuit receiving the intervention (b). Diamonds and intervals summarize trained models; faint paths show individual cells.

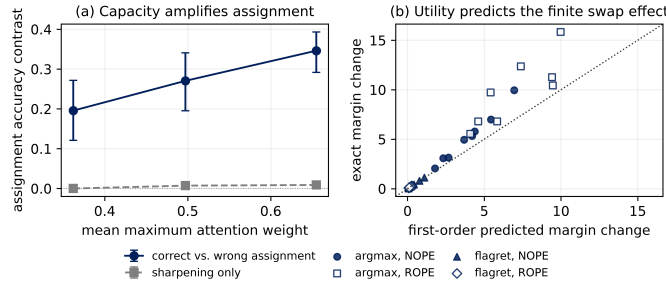


Figure 4: Capacity controls leverage, and utility predicts the finite intervention. Sharper spectra increase the correct-versus-wrong assignment contrast while sharpening alone remains near zero (a). The first-order utility term predicts the exact source-max margin effect across tasks and positional encodings (b). Error bars summarize trained models.

5 Results and Discussion

5.1 Assignment Changes Accuracy at Fixed Spectrum

We begin with the simplest test: move the largest available weight to the source, move it away, or exchange two distractors. Across sixteen trained models, the mean changes are $+0.153$, -0.043 , and $+0.011$, respectively (Figure 3). Source-max beats the control in every model-length comparison. Their primary cluster-level contrast is $+0.142$ (95% CI $[+0.080, +0.206]$; exact $p = 3.05 \times 10^{-5}$; Holm-adjusted $p = 9.16 \times 10^{-5}$), while complete-spectrum invariant error stays below 10^{-6} .

The effect grows with circuit coverage (Figure 3b): patching one, two, four, and eight selected heads gives mean source-max gains of 0.007, 0.013, 0.056, and 0.153, monotone in 31 of 32 comparisons, consistent with a distributed effect. Source attention tracks accuracy more closely than attention-output variance: normalizing that variance leaves accuracy unchanged, whereas forcing mass onto the known source restores controlled retrieval, and sharpening the original row has little effect. The ordering survives the eight-layer replication, both positional encodings, four seeds, and both controlled tasks, while random-layer and adjacent-layer interventions stay near zero. The full displacement audit, per-cell intervals, and variance and patching results are in the supplementary material.

5.2 Capacity Amplifies Assignment

We next vary spectrum and assignment independently. Temperature scaling changes the maximum available weight, and at every temperature we evaluate identity, source-max, source-min, and distractor control on the same examples. The source-max minus source-min contrast grows from 0.196 to 0.346, while sharpening the identity assignment changes accuracy by at most 0.009 (Figure 4a). Capacity sets the possible leverage, while assignment determines where that leverage lands. The

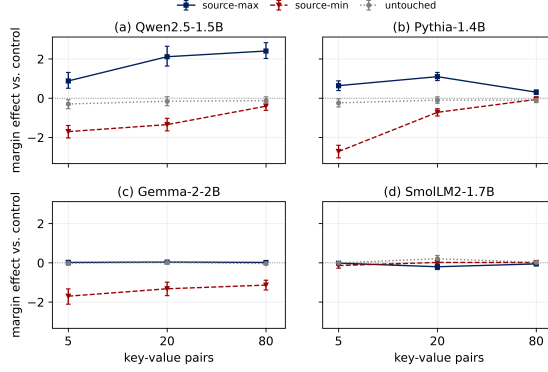


Figure 5: Matched pretrained-model comparison. A single calibration seed (seed 0) is shown because the frozen circuit is selected once per seed and the panels compare conditions within that fixed circuit; seed-pooled results are in the supplementary material. Error bars are paired 95% intervals over $n = 128$ examples per model-length cell. All panels use the 5-, 20-, and 80-pair ladder and report source-max, source-min, and untouched margin relative to the same distractor control. Qwen and Pythia respond positively, Gemma is saturated, and the source-mass-selected SmoLLM2 circuit does not benefit.

primary interaction is $+0.150$ (95% CI $[+0.094, +0.210]$; exact $p = 6.10 \times 10^{-5}$; Holm-adjusted $p = 1.22 \times 10^{-4}$).

For held-out examples we differentiate the target margin along the exact source-max path. The first-order term in Proposition 2 ranks the finite margin change, with per-model Spearman correlations of 0.87, 0.54, and 0.55 on Qwen2.5-1.5B, Pythia-1.4B, and Gemma-2-2B and sign agreements of 0.81, 0.81, and 0.71 (Appendix A). Aggregating to the 32 controlled cell means in Figure 4b raises the rank correlation to 0.989, but that pooled figure overstates the per-row relation and we do not use it as the headline. The slope of the actual-on-predicted fit is 1.387, off by about 39%, so we read the utility term as a usable local ranking signal rather than a calibrated estimator of effect size. The remaining error measures downstream curvature along the swap.

Modular sum requires the model to route and combine two marked values. At $5 \times$ training length, assigning the top two weights to evidence raises exact match by 0.066, assigning the bottom two lowers it by 0.291, and evidence-max exceeds a control that preserves both the spectrum and total evidence mass by 0.043 (95% CI $[0.017, 0.072]$). This is a small but positive two-evidence effect at $5 \times$ (the effect at $1 \times$ is $+0.007$), so the result reaches two required tokens rather than one; it does not establish general multi-token routing. A registered three-evidence estimate includes zero, and four-evidence models miss the frozen competence gate, so arity beyond two remains unresolved.

Evidence-max and evidence-min preserve the complete spectrum. The multi-evidence control also preserves total evidence mass, changing only which evidence token receives each weight. Its small effect separates aggregate source mass from useful evidence assignment.

5.3 The Same Intervention Produces Three Pretrained Regimes

The pretrained models show why the utility term matters. Qwen and Pythia gain margin when source-max replaces the distractor control. Gemma stays accurate as context grows and gains little from extra source mass, even though source-min reliably harms it. This is the saturation regime predicted by Proposition 2: removing evidence matters, while extra routing has little left to add. A quantized Qwen2.5-7B run preserves the positive margin direction.

SmoLLM2 provides the contrasting case. Its preregistered source-mass selector fails, yet ranking heads on disjoint calibration examples by output-conditioned gradient or utility restores a positive effect. A ten-seed ablation ranks source gradient, utility gain, and utility gap above source mass and transfer mass. Rankings vary by architecture, but circuit selection must account for the output.

Interpolating from the natural row to the utility-selected swap gives a monotone margin path over five calibration seeds. On context-grounded natural QA, utility-selected routing improves margin over

Task	Cond.	Unpatched	Δ src-max	Δ src-min	Δ distr.	max—distr.
Single evid. (Acc)	$\beta = 1$	.619	+ .153	− .043	+ .011	+ .142
	$\beta = 4$	.619	+ .298	− .048	+ .025	+ .273
Two evid. (EM)	$1 \times$	.952	+ .019	− .559	+ .012	+ .007
	$5 \times$	.484	+ .066	− .291	+ .023	+ .043

Table 1: Controlled fixed-spectrum assignment. Single-evidence rows report per-token accuracy over $n = 8,192$ head-example pairs; two-evidence rows report exact match over $n = 2,048$. Source-max moves the row maximum to the source, source-min moves it away, and the distractor control swaps two non-source weights; all preserve the complete spectrum. Full cells and 95% intervals are in the supplementary material.

Model	Unpatched	Δ src-max	Δ src-min	max—distr.
Qwen2.5-1.5B	−3.048	+1.990	−.961	+1.799
Pythia-1.4B	−3.805	+ .826	−1.022	+ .680
Gemma-2-2B	+2.234	+ .040	−1.373	+ .027
SmolLM2-1.7B	−2.782	−.168	−.112	−.091
<i>Output-conditioned circuit selection (ΔM vs. matched control)</i>				
	util. gain	util. gap	src mass	util—src
Pythia-1.4B	+ .519	+ .359	+ .006	+ .513
SmolLM2-1.7B	+1.253	+ .859	+ .006	+1.247
Qwen2.5-1.5B	+1.493	+1.493	+ .069	+1.424

Table 2: Pretrained models, all in target-margin units ($n = 384$ per model, matched 5/20/80-pair ladder from Figure 5). Top: fixed-spectrum assignment. Bottom: output-conditioned circuit selection, where selecting by source mass gives near-zero margin change (src mass column) while selecting by a value-aware utility term is large and positive (util—src column); a Gemma selector run was not conducted. Full cells and 95% intervals are in the supplementary material.

a displacement-matched control, while source-mass selection harms it. The direction is consistent across all six disjoint calibration seeds (mean contrast +.378), but the magnitude is not evenly distributed: one seed carries most of the pooled mean, and a leave-one-seed-out check ranges from +.19 to +.55, so we read the result as directionally reliable and magnitude-uncertain. We deliberately avoid a significance claim here: an exact two-sided sign test at six seeds has a minimum attainable p of .03125, which we reach, so the test has no resolution to distinguish this effect from any all-positive one and the interval, not a p -value, is the honest summary. All competence gates pass, and an unconstrained generation audit finds no first-answer or repetition cost.

The selector results judge a selector by the frozen circuits it finds and their paired margin effects. They do not test whether the per-row utility functional predicts a finite swap effect. That held-out predictor audit is reported separately in the supplementary material.

5.4 Two results that bound the causal claim

Two measurements limit how far the fixed-spectrum effect should be read as the model’s own mechanism, and we state them here rather than as asides.

First, the effect is measured on rows the model does not itself produce. On untouched distractor-order variants, where we do not swap, the conditional source-mass coefficient is +1.250 ($p = .03125$), so source mass is associated with the outcome in endogenous data. The matched fixed-spectrum contrast on those same endogenous rows, however, is +.026 with a 95% interval of [−.017, +.070] that includes zero. The intervention thus measures sensitivity to a counterfactual reassignment, and it does not establish that normal query-key computation creates the swapped rows. This is the central caveat of an off-manifold probe, and the near-zero endogenous contrast is why we frame FSS as a test of a specified reassignment rather than of the model’s spontaneous routing.

Second, restoring the selected circuit recovers little of the damage from corrupting it. Clean-to-corrupt activation patching on the selected heads rescues +.308 margin, which is only 2.5 to 3.3% of the

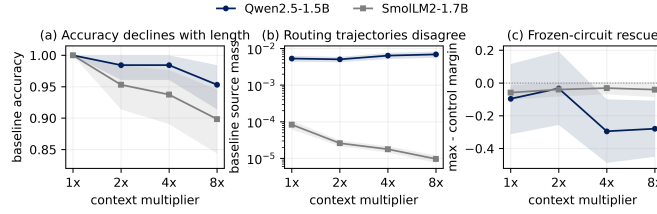


Figure 6: Matched natural-QA length ladder (seed 0; the four-passage circuit is frozen once per seed and reused at every length, and the additional Qwen seed is in the supplementary material). Bands are paired bootstrap 95% intervals over $n = 128$ examples per cell. From 4 to 32 passages, accuracy falls (a), source routing moves in opposite directions across models (b), and frozen-circuit rescue stays flat or weakens (c).

median corruption damage. Because patching restores head outputs after source-value corruption while reassignment acts on an unchanged input, the two magnitudes are not directly comparable; still, the small recovered fraction indicates that the selected heads are not where most of the causal action sits, and the effect we report is one contribution among several rather than the dominant pathway.

5.5 Natural Context Marks the Scope Boundary

The fixed-context natural-QA result establishes causal routing in ordinary text. The length ladder is a scope test: we select the circuit at four passages, freeze every choice, and append unrelated passages to form nested 4-, 8-, 16-, and 32-passage contexts. Both Qwen and SmolLM2 lose margin and accuracy by 32 passages, yet their source-mass trajectories diverge: Qwen source mass rises as rescue weakens, while SmolLM2 source mass falls with rescue flat. The shared degradation cannot be attributed to one consistent source-mass trajectory, so this is a scope limit of the diagnosis, not a failed prediction. At a fixed context the swap measures whether a chosen circuit can use reassigned evidence; as contexts grow, representation quality, distractor semantics, and computation outside the circuit also matter, and the theorem, which concerns a specified swap at a fixed context, does not resolve those bottlenecks. Exact 32-minus-4 changes are in the supplementary material.

5.6 Identification and Replication Audits

Replication checks vary the calibration sample, patch budget, and model family, and they support the same reading. Selected circuits are positive at every head budget, adjacent and random layers are smaller or near zero, the utility term ranks exact effects within every competent cell while calibrating them imperfectly, and an exact rerun reproduces the direction with an atypically large original magnitude. Appendix A reports these audits in full.

6 Limitations

FSS identifies the causal effect of a specified post-softmax reassignment, not a complete account of model computation: the swap is an off-manifold counterfactual, and preserving the spectrum isolates assignment without showing that training produces the same row. One control is also missing and bounds the natural-QA reading. The displacement-matched control moves mass between two task-irrelevant tokens, whereas the treatment moves it onto a semantically loaded token; a stronger control moves the same maximum onto a competitor answer token, matched for spectrum and displacement, to separate assignment to required evidence from assignment to any answer-shaped token (source-min is not a substitute, since it also changes the magnitude moved). We flag this semantic-distractor control as the primary missing comparison. Finally, higher arity is competence-limited, the generation audit is short-horizon, the six-seed families have no resolution below the sign-test floor and are reported descriptively, and utility ranks swaps better than it calibrates them. Anonymized code, seeds, prompts, and per-example outputs accompany the submission and will be released on acceptance.

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352 A Identification and replication audits

353 Replication checks vary the calibration sample, patch budget, and model family. Across three frozen
 354 Qwen calibration seeds, selected circuits are positive at every head budget: $K = 2, 4, 8$ gives $+ .460$,
 355 $+1.054$, and $+1.107$ margin. Adjacent layers are smaller and inconsistent ($+ .147$ and $- .131$), while
 356 random layers average near zero ($- .059$). Full intervals and seed standard deviations are reported in
 357 the supplementary package.

358 A separate replication tests the utility relation. Circuits are chosen by source mass on disjoint
 359 calibration examples; utility is then computed on held-out evaluation rows. Across three seeds per
 360 family, every competent source-max cell is positive, and predicted effects rank the exact effects
 361 within each cell. Mean Spearman correlations are $.866$ for Qwen2.5-1.5B, $.544$ for Pythia-1.4B, and
 362 $.554$ for Gemma-2-2B, with sign agreements $.805$, $.807$, and $.714$. The approximation is imperfect at
 363 large effects, so we use it as a local ranking signal rather than a calibrated finite-effect estimator. For
 364 the 32 controlled cell means in Figure 4b, the actual-on-predicted fit is $1.387x - .102$ (Pearson $.978$),
 365 which deviates from identity calibration.

366 The exact Qwen rerun keeps precision, lengths, contrast, and sample counts fixed. It is the clearest
 367 case of directional consistency with magnitude instability: the original seed gives $+1.882$, the five
 368 new seeds give a mean of $+ .042$, and all six are positive. The sign is reproducible while the magnitude
 369 is not, and the pooled mean of the natural-QA contrast in Section 5.3 is dominated by the original
 370 seed. Because an exact two-sided sign test at six seeds bottoms out at $p = .03125$, we report the
 371 seed-level values and the leave-one-seed-out range rather than a significance verdict for these six-seed
 372 families. Adding seeds is the natural next step and is left to an extended version.

373 Output-conditioned selectors lead in each family, although their internal ranking changes by architec-
 374 ture. Full selector results, competence-gate failures, and the anonymized code package appear in the
 375 supplementary material.

376 B Theory details

377 This appendix gives the full statements and proofs behind Propositions 1 and 2, and a multi-evidence
 378 corollary. The complete per-cell result tables and secondary figures are in the separate supplementary
 379 material.

380 **Established softmax grouping identity.** For evidence positions S , distractors D , scores s_j , and
 381 temperature $T > 0$, let $Z_G = \sum_{j \in G} e^{s_j/T}$ and $F_G = -T \log Z_G$. Grouping the softmax denomina-
 382 tor gives

$$m_S := \sum_{j \in S} a_j = \frac{Z_S}{Z_S + Z_D} = \sigma\left(\frac{F_D - F_S}{T}\right). \quad (7)$$

383 With the group log-mean-exp score $\bar{s}_G = T \log(|G|^{-1} \sum_{j \in G} e^{s_j/T})$,

$$m_S \geq \alpha \iff \bar{s}_S - \bar{s}_D \geq T \left(\log \frac{|D|}{|S|} + \text{logit}(\alpha) \right). \quad (8)$$

384 These Gibbs partition-function identities [Ramsauer et al., 2021] make the dependence on distractor
385 scores explicit: distractor count alone is insufficient, since arbitrarily many negligible-score distractors
386 need not reduce evidence mass.

387 **Capacity–assignment decomposition.** Let $a_{(1)} \geq \dots \geq a_{(n)}$, let S contain r evidence positions,
388 and define $C_r^+ = \sum_{i=1}^r a_{(i)}$, $C_r^- = \sum_{i=n-r+1}^n a_{(i)}$, and $K_r(a) = C_r^+ - C_r^-$.

389 **Theorem 1** (Capacity–assignment decomposition). *For every permutation π of a fixed attention-*
390 *weight multiset, $C_r^- \leq m_S(\pi) \leq C_r^+$, both endpoints are attainable, and for $K_r(a) > 0$ the unique*
391 *coordinate $\rho_S(\pi) = (m_S(\pi) - C_r^-)/K_r(a) \in [0, 1]$ gives $m_S(\pi) = C_r^- + \rho_S(\pi)K_r(a)$. For affine*
392 *margin $M(\pi) = b + \sum_j a_{\pi(j)}u_j$ with sorted utilities $u_{(1)} \geq \dots \geq u_{(n)}$,*

$$\max_{\pi} M(\pi) = b + \sum_i a_{(i)}u_{(i)}, \quad \min_{\pi} M(\pi) = b + \sum_i a_{(i)}u_{(n+1-i)}. \quad (9)$$

393 *No statistic invariant to token permutation can determine evidence assignment or the output margin.*

394 *Proof.* An r -position set receives at most the r largest weights and at least the r smallest; assigning
395 those weights to S attains the endpoints. Normalizing the bounded mass gives ρ_S . For the output
396 range, suppose $u_p > u_q$ but $a_{\pi(p)} < a_{\pi(q)}$. Swapping those weights changes the margin by
397 $(a_{\pi(q)} - a_{\pi(p)})(u_p - u_q) > 0$. Repeated exchanges align weights and utilities for the maximum
398 and reverse them for the minimum, the classical rearrangement inequality [Hardy et al., 1952]. The
399 extremal assignments preserve the complete weight multiset and therefore every permutation-invariant
400 statistic. $\square$

401 **Nonlinear downstream network.** Fix an input and selected heads H at one layer. Let $x_{hj} =$
402 $O_h v_{hj}$ and $z = \sum_{h \in H} \sum_j a_{hj} x_{hj}$. For target y and competitor c , let $M_c(z) = g_y(z) - g_c(z)$. A
403 source–donor swap has $\delta_h = a_{hd_h} - a_{hs}$ and $\Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h})$. Define local utility
404 $u_{hj}^{(c)} = \nabla M_c(z)^\top x_{hj}$.

405 **Theorem 2** (Nonlinear routing-margin bound). *If ∇M_c is L_c -Lipschitz from z to $z + \Delta z$, then*

$$\left| \Delta M_c - \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) \right| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \quad (10)$$

406

407 *Proof.* The swap changes head h by exactly $\delta_h (x_{hs} - x_{hd_h})$. The fundamental theorem of calculus
408 gives $\Delta M_c = \int_0^1 \nabla M_c(z + t\Delta z)^\top \Delta z dt$. Subtracting $\nabla M_c(z)^\top \Delta z$ and applying Lipschitzness
409 bounds the remainder by $\int_0^1 L_c t \|\Delta z\|_2^2 dt = L_c \|\Delta z\|_2^2 / 2$ [Nesterov, 2004]. Expanding the first-order
410 term gives Equation 10. $\square$

411 **Corollary 3** (Multi-evidence routing). *Let S contain r evidence positions and pair every evidence*
412 *position missing a top- r weight with one top- r non-evidence donor. For disjoint swap pairs P_h and*
413 *$\delta_{hsd} = a_{hd} - a_{hs} \geq 0$, $\Delta z = \sum_{h \in H} \sum_{(s,d) \in P_h} \delta_{hsd} (x_{hs} - x_{hd})$, and Equation 10 holds with the*
414 *first-order sum extended over $(s, d) \in P_h$. Assigning top- r capacity to the evidence improves the*
415 *margin whenever the summed evidence–donor utility gain exceeds the curvature remainder.*

416 The decomposition and corollary are algebraic for arbitrary r . Their behavioral premise is established
417 for $r = 1$ and $r = 2$; the three-evidence interval includes zero and the four-evidence models miss the
418 competence gate, so efficacy beyond $r = 2$ remains open. The condition is sufficient, not necessary:
419 source-max can fail when evidence values have no greater downstream utility, patched heads cancel,
420 or curvature dominates, so natural source mass alone is not a universal circuit-selection rule.

C Experimental protocol

Controlled tasks and training. Argmax retrieval requests the value paired with the largest key; flag retrieval requests the single marked value; both are order-invariant with one identifiable evidence position. Reversed decimal addition is an order-dependent contrast, and two-evidence modular sum marks two digits and requests their sum modulo ten so copying either digit is insufficient. The primary model is a four-layer pre-norm decoder transformer with width 256, eight heads, and feed-forward width 1024, trained with AdamW, batch size 512, and warmup plus cosine decay on lengths one through five. Evaluation follows a geometric ladder to $50\times$ training length, and the scale replication uses eight layers and width 512. A cell is interpreted only after training-length exact match reaches 0.8. Order-invariant cells use four seeds; the order-dependent contrast uses two. The preregistration fixes the competence gate, break-point rule, and variance-collapse check before intervention outcomes.

Measured quantities and interventions. At the answer query we record evidence mass, attention entropy, every attention norm, participation ratio, attention-output variance, and target margins. Source-max swaps the source weight with the row maximum, source-min swaps it with the row minimum, and the distractor control swaps the largest and smallest non-source weights. All conditions preserve model parameters, prompts, transformed values, the complete sorted attention row, entropy, norms, maximum, and participation ratio; the control additionally preserves source mass.

Confirmatory inference. One family of three primary contrasts is fixed in advance: controlled source-max minus distractor control, the capacity-by-assignment interaction, and natural utility-selected source-max minus matched control. Exact two-sided sign-flip tests operate on independent trained-model clusters for the controlled claims and calibration-seed clusters for natural QA, and Holm adjustment controls familywise error across the three tests. Hierarchical intervals and all other reported cells are descriptive and unadjusted unless stated otherwise.

D Scope, boundaries, and reproducibility

Scope. The equals-sign second-prompt format fails its competence gate in Qwen, and a blinded arrow-format pilot fails full-run competence despite directional margin signs; both are excluded from causal replications. Pythia and Qwen are floor-limited at their longest contexts, so margin is the interpretable tail outcome; Qwen-7B is quantized, and Gemma supplies a high-competence saturation boundary. The fixed-spectrum theorem still applies to the natural-QA length result because it conditions effects on transferred mass, value utility, and curvature, but the length ladder does not support the stronger claim that natural length degradation must involve source-mass loss.

Reproducibility. The controlled tasks are generated from the included task generator and require no external dataset. Natural MCQA samples are constructed from the public SQuAD training split [Rajpurkar et al., 2016] with the sampling seed and eligibility rule included. Pretrained probes use publicly released Qwen, Pythia, Gemma, and SmolLM2 checkpoints, with exact identifiers, tokenizer settings, and precision recorded in the result artifacts. Pretrained experiments used a Google Colab runtime with one NVIDIA T4 GPU, PyTorch, and Hugging Face Transformers; the earlier controlled artifacts retain configurations and seeds but not the exact package-version snapshot, so we mark infrastructure reproducibility partial. Saved artifacts reproduce every reported statistic and figure without loading a language model, and an anonymized code and artifact archive accompanies the submission as supplementary material.